

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A model estimates that whales from the genus *Eschrichtius* travel **72** to **77** miles in the ocean each day during their migration. Based on this model, which inequality represents the estimated total number of miles, x , a whale from the genus *Eschrichtius* could travel in **16** days of its migration?

Answer

- A. $72 + 16 \leq x \leq 77 + 16$
- B. $(72)(16) \leq x \leq (77)(16)$
- C. $72 \leq 16 + x \leq 77$
- D. $72 \leq 16x \leq 77$

Correct Answer: B

Rationale

Choice B is correct. It's given that the model estimates that whales from the genus *Eschrichtius* travel **72** to **77** miles in the ocean each day during their migration. If one of these whales travels **72** miles each day for **16** days, then the whale travels **72(16)** miles total. If one of these whales travels **77** miles each day for **16** days, then the whale travels **77(16)** miles total. Therefore, the model estimates that in **16** days of its migration, a whale from the genus *Eschrichtius* could travel at least **72(16)** and at most **77(16)** miles total. Thus, the inequality $(72)(16) \leq x \leq (77)(16)$ represents the estimated total number of miles, x , a whale from the genus *Eschrichtius* could travel in **16** days of its migration.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.